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Course: COP1034C Programming for Technology Professionals

Car Evaluation Dataset

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Data information

Derived from a simple hierarchical decision model, this dataset may be useful for testing constructive induction and structure discovery methods:

- **buying:** vhigh, high, med, low.
- **maint:** vhigh, high, med, low.
- **doors:** 2, 3, 4, 5.
- **persons:** 2, 4, 5.
- **lug_boot:** small, med, big.
- **safety:** low, med, high.
- **class:** unacc, acc, good, vgood

Source: [Car Evaluation](#) on UCI Machine Learning Repository

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1727 entries, 0 to 1726
Data columns (total 7 columns):
#   Column      Non-Null Count  Dtype
---  -
0   vhigh       1727 non-null   object
1   vhigh.1     1727 non-null   object
2   2           1727 non-null   object
3   2.1         1727 non-null   object
4   small       1727 non-null   object
5   low         1727 non-null   object
6   unacc       1727 non-null   object
dtypes: object(7)
memory usage: 94.6+ KB

```

Figure 1 – The information of the data

Analysis:

- Missing Data: There are no missing values in any of the columns, as indicated by the non-null count matching the total number of entries (1727) for each column.
- Data Types: The categorical variables are represented as object type.
- Memory Usage: The dataset uses 94.6 KB of memory, which is manageable for most analyses.

Overall:

- The dataset is clean with no missing values, and the data types are relatively appropriate for the categorical nature of the variables. However, the columns should be renamed for clarity would be beneficial and the datatypes of doors and persons should be changed to facilitate subsequent analysis.

Data size and dimension

Dataset shape: (1727, 7)

	buying	maint	doors	persons	lug_boot	safety	class
0	vhigh	high	2	2	small	med	unacc
1	vhigh	vhigh	2	2	small	high	unacc
2	vhigh	high	2	2	med	low	unacc
3	vhigh	vhigh	2	2	med	med	unacc
4	vhigh	vhigh	2	2	med	high	unacc

Table 1 - The first 5 rows of the dataset.

This dataset consists of 1727 rows and 7 columns. Each row represents a unique car evaluation, and each column corresponds to a specific attribute or feature of the car. The columns include:

1. **buying:** Buying price of the car (Categorical)
2. **maint:** Maintenance price of the car (Categorical)
3. **doors:** Number of doors in the car (Numerical)
4. **persons:** Capacity in terms of persons to carry (Numerical)
5. **lug_boot:** Size of the luggage boot (Categorical)
6. **safety:** Estimated safety of the car (Categorical)
7. **class:** Evaluation level of the car (Target variable, Categorical)

Missing data

	0
buying	0

maint	0
doors	0
persons	0
lug_boot	0
safety	0
class	0

Table 2 – Missing data counts

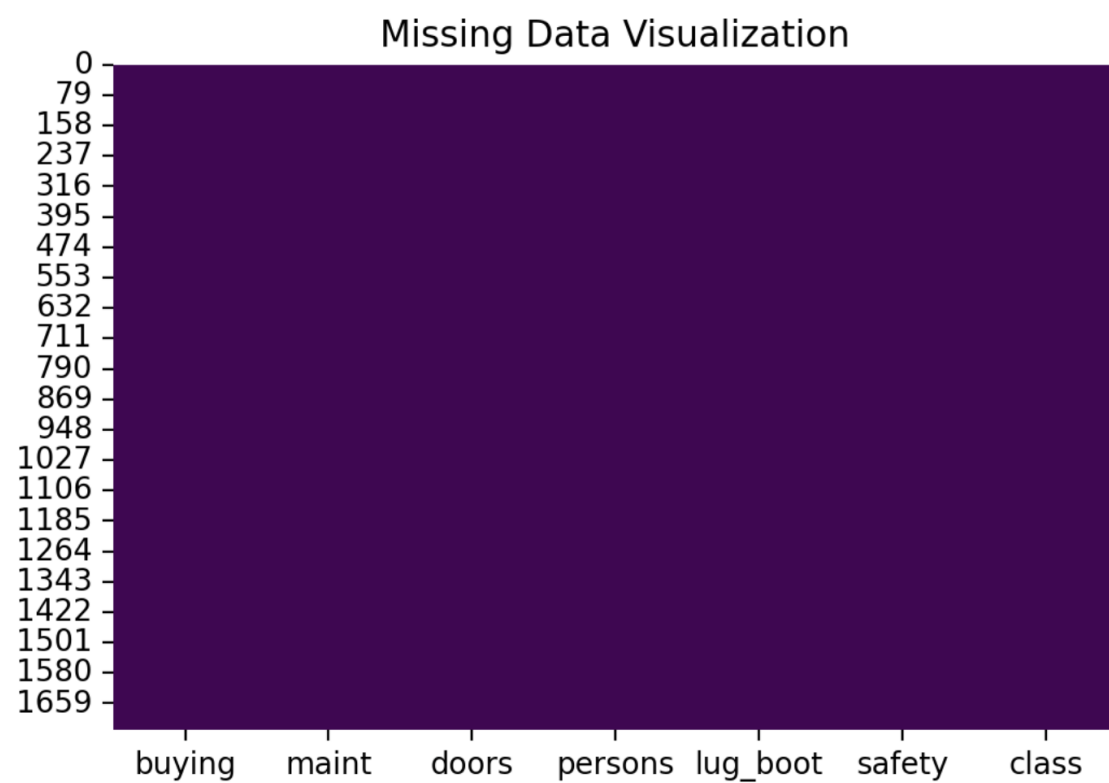


Figure 2 – Missing Data Visualization

- The analysis of missing data in the dataset reveals that there are no missing values across all columns. This is shown by the missing data count table, which shows zero missing values for each column.
- The heatmap visualization further supports this finding, as it does not display any gaps or missing data points. This also indicates that the dataset is complete and does not require any imputation or handling of missing values.

Duplicated values

No duplicated rows found.

Number of Duplicated Rows: 0

- The analysis of duplicate data in the dataset indicates that there are no duplicate rows. This is confirmed by the output above, which shows zero duplicated rows.

Numerical data columns

	doors	persons
count	1,727	1,727
mean	3.5009	3.6676
std	1.1181	1.2473
min	2	2
25%	3	2
50%	3	4
75%	4.5	5

max	5	5
-----	---	---

Table 3 – Summary Statistic of the numerical columns

Analysis:

- **Numeric Columns:** `doors` and `persons` are the two numeric columns in the dataset.
- **Central Tendency:** The mean and median values for both `doors` (at 3.5) and `persons` (at 3.67) are close to each other, indicating a relatively symmetric distribution for these columns.
- **Dispersion:** The standard deviation values indicate the spread of the data around the mean. For `doors`, the standard deviation is 1.118, and for `persons`, it is 1.248. This suggests that the number of doors and persons varies moderately around their respective means.
- **Range:** The range (difference between the minimum and maximum values) for both columns is 3, indicating a moderate spread of values.
- **Percentiles:** The percentiles provide insights into the distribution of the data. For `doors`, 25% of the data has 2.75 or fewer doors, 50% has 3.50 or fewer, and 75% has 4.25 or fewer; and for `persons`, 25% of the data can accommodate 2 or fewer persons, 50% can accommodate 4 or fewer, and 75% can accommodate 5 or fewer.
- **Outliers:** The minimum and maximum values for both columns are within a reasonable range, suggesting that there are no extreme outliers.

Categorical data columns

In this dataset, there are 5 columns categorized as categorical variables. These columns represent distinct groups or categories of information, rather than continuous numerical data. The categorical data columns are as follows:

1. buying: ['vhigh', 'high', 'med', 'low']
2. maint: ['vhigh', 'high', 'med', 'low']
3. lug_boot: ['small', 'med', 'big']
4. safety: ['low', 'med', 'high']
5. class: ['unacc', 'acc', 'vgood', 'good']

Analysis

Closeness of Data:

- The unique values in each categorical column are well-defined and distinct. This indicates that the categories are clearly separated and there is no overlap between them. Additionally, each unique value all has enough data.

Abnormalities:

- There are no apparent abnormalities in the unique values of the categorical columns. Each column has a reasonable number of unique values that are expected for the type of data being analyzed.

Outliers:

- Based on the unique values provided, there are no immediate outliers.

Overall

- The data of categorical values does not look unusual. Each unique value has complete data and no outliers. The data distribution of values will be shown more clearly in the next steps.

Plotting Distribution and Box Plots for Numeric Columns

Distribution of 'Doors' Variable

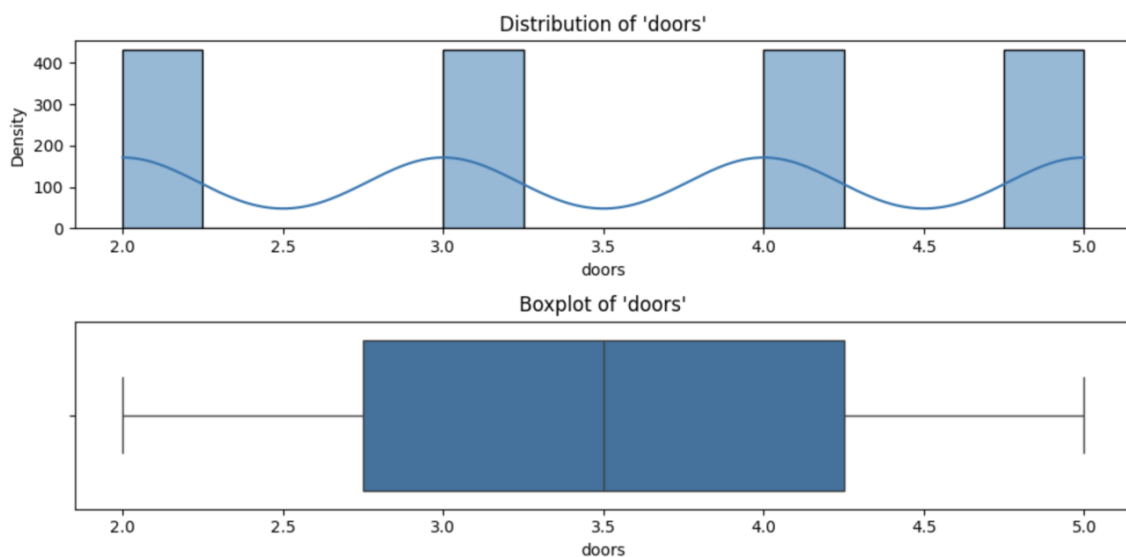


Figure 3 – Plotting Distribution and Box Plot 'doors' Columns

Histogram and Density Plot

The first plot in the image is a histogram with a density plot overlay. It provides a visual summary of the distribution of the 'doors' variable:

- X-Axis: Represents the number of doors, ranging from 2 to 5.
- Y-Axis: Represents the density, ranging from 0 to 400.
- Data values: [2.0, 3.0, 4.0, 5.0]
- Histogram Bars: There are four bars at 2, 3, 4, and 5 doors, each with a density of approximately 400. This indicates that these four categories have a similar frequency.
- Density Plot: A smooth curve that oscillates, peaking at the same points as the histogram bars. This provides a smoothed estimate of the distribution, helping to understand the shape of the data.

Boxplot

The second plot is a boxplot of the 'doors' variable, which provides insights into the central tendency and variability:

- X-Axis: Represents the number of doors, ranging from 2 to 5.
- Interquartile Range (IQR): Spans from approximately 2 to 4 doors, indicating the middle 50% of the data.
- Median: Located around 3 doors, indicating that half of the cars have 3 or fewer doors.
- Whiskers: Extend from 2 to 5 doors, representing the range of the data. There are no apparent outliers.

Relevance

- Symmetric Distribution: Both the histogram and the density plot suggest a relatively symmetric distribution for the number of doors.
- Moderate Variability: The boxplot shows that the number of doors varies moderately, with most data points falling within the interquartile range.
- No Extreme Outliers: The absence of outliers in the boxplot suggests that the data for the number of doors is well-contained within a reasonable range.

Overall

- These plots provide a comprehensive overview of the distribution and variability of the 'doors' variable, aiding in the understanding of this key attribute in the dataset.

Distribution of 'Persons' Variable

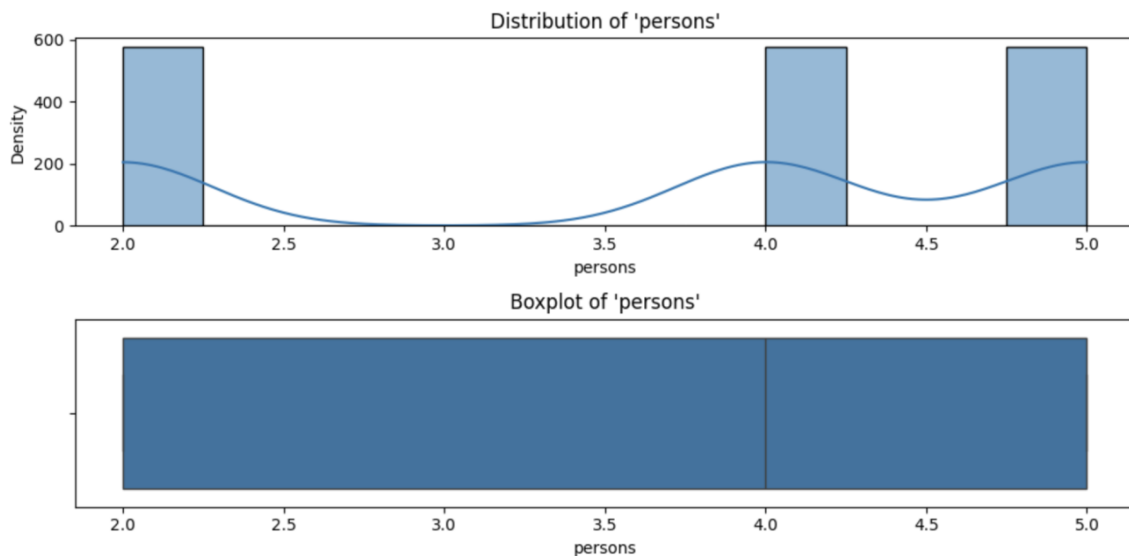


Figure 4 – Plotting Distribution and Box Plot 'persons' Columns

Histogram and Density Plot

The first plot is a histogram with a density plot overlay, providing a visual summary of the distribution of the 'persons' variable:

- X-Axis: Labeled 'persons', with values ranging from 2.0 to 5.0.
- Y-Axis: Labeled 'Density', ranging from 0 to 600.
- Data values: [2.0, 4.0, 5.0]
- Histogram Bars: Three distinct bars at values 2.0, 4.0, and 5.0, each with a height of approximately 600, and missing data at value 3.0.
- Density Plot: A smooth curve that peaks at 2.0, 4.0, and 5.0, aligning with the histogram bars. This suggests fewer occurrences of values between these peaks.

Boxplot

The second plot is a boxplot of the 'persons' variable, summarizing the central tendency and variability:

- X-Axis: Labeled 'persons', ranging from 2 to 5.
- Interquartile Range (IQR): Spans from 2.0 to 5.0, the median is at 4.0 because the data only has 3 values, so the median is skewed.
- Median: Around 4.0, indicating half the cars can accommodate 4 or fewer people.
- Whiskers: Extend from 2.0 to 5.0, covering the full range of data without apparent outliers.

Relevance

- **Frequent Values:** Both plots highlight that 2.0, 4.0, and 5.0 are the most frequent values for 'persons', and data deficiency at value 3.0.
- **Symmetric Distribution:** The data of 'persons' is not symmetric because it only consists of 3 values.
- **Moderate Variability:** The standard deviation values indicate the data spreads moderately around the mean.
- **No Extreme Outliers:** The absence of outliers in the boxplot suggests the data is well-contained within a reasonable range.

Overall

- Together, these plots provide a comprehensive overview of the 'persons' variable, aiding in understanding its distribution and central tendency in the dataset.

Plotting Horizontal Bar Plots for Categorical Columns

Analysis of 'Buying' Variable

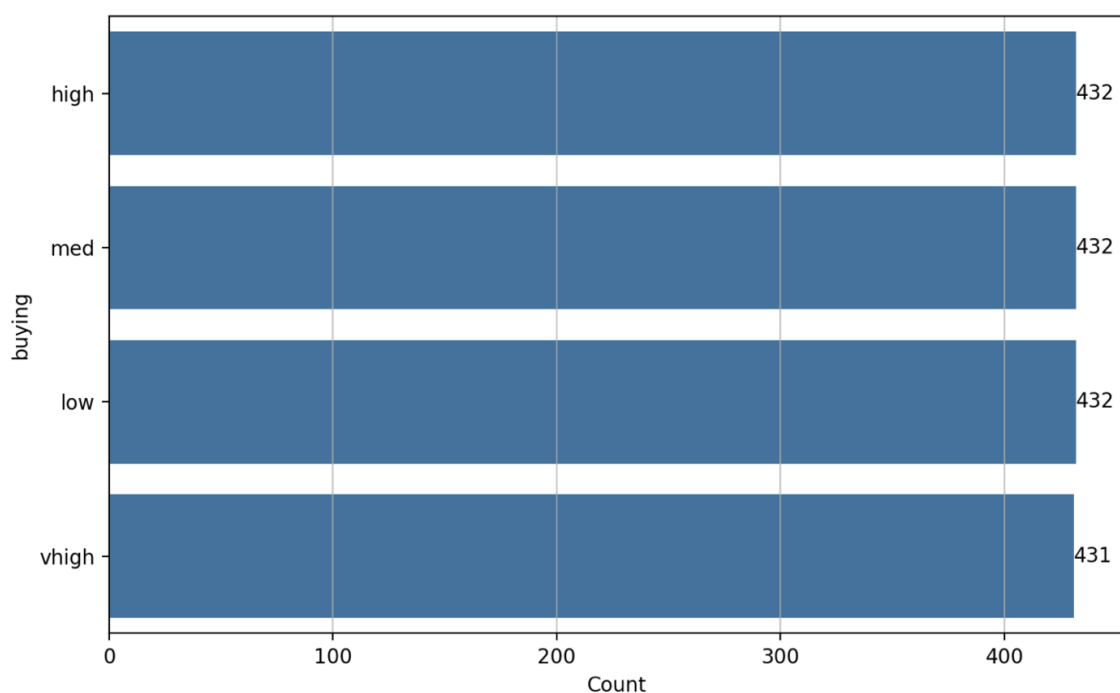


Figure 5 – Plotting Horizontal Bar Plots for 'buying' Columns

The horizontal bar chart provides a visual summary of the distribution of the 'buying' variable, which represents the buying price categories of the cars in the dataset.

- Categories: The 'buying' variable includes four categories "vhigh" (very high), "high," "med" (medium), and "low."
- Count: the "high," "med" (medium), and "low" have an equal count of 432, the count of "vhigh" (very high) is lower than the three categories above at 431. The x-axis displays the count, ranging from 0 to 400, while the y-axis lists the categories.

Analysis of 'maint' Variable

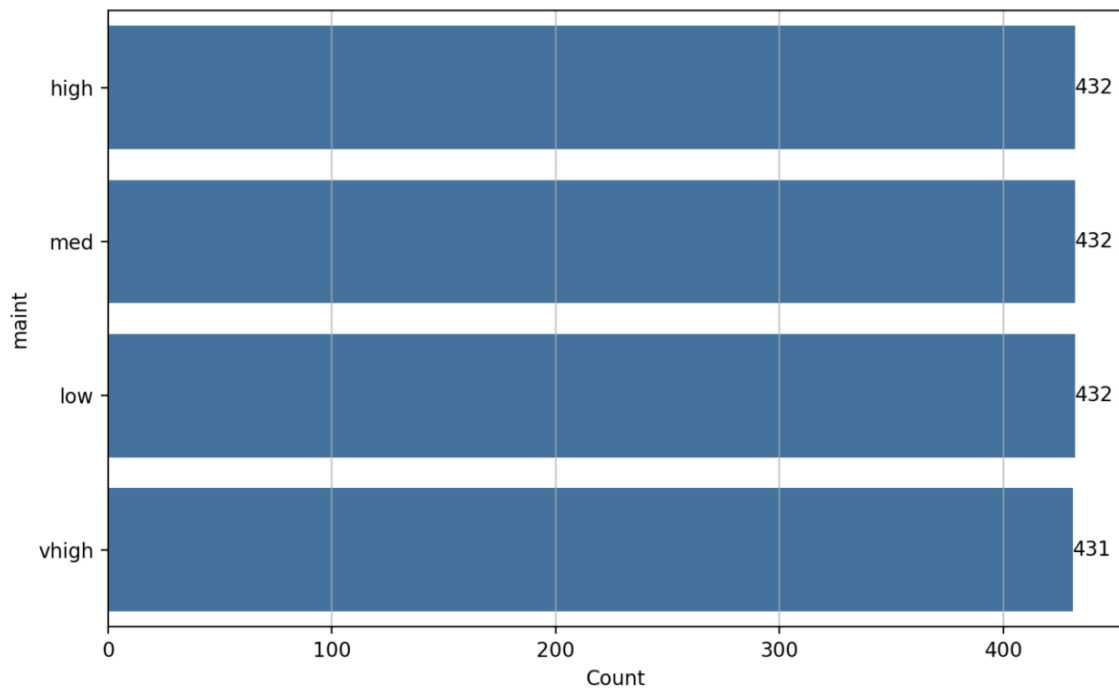


Figure 6 – Plotting Horizontal Bar Plots for 'maint' Columns

The horizontal bar chart provides a visual summary of the distribution of the 'maint' variable, which represents the maintenance price of the cars in the dataset.

- Categories: The 'maint' variable includes four categories "vhigh" (very high), "high," "med" (medium), and "low."
- Count: the "high," "med" (medium), and "low" have an equal count of 432, the count of "vhigh" (very high) is lower than the three categories above at 431. The x-axis displays the count, ranging from 0 to 400, while the y-axis lists the categories.

Analysis of 'lug_boot' Variable

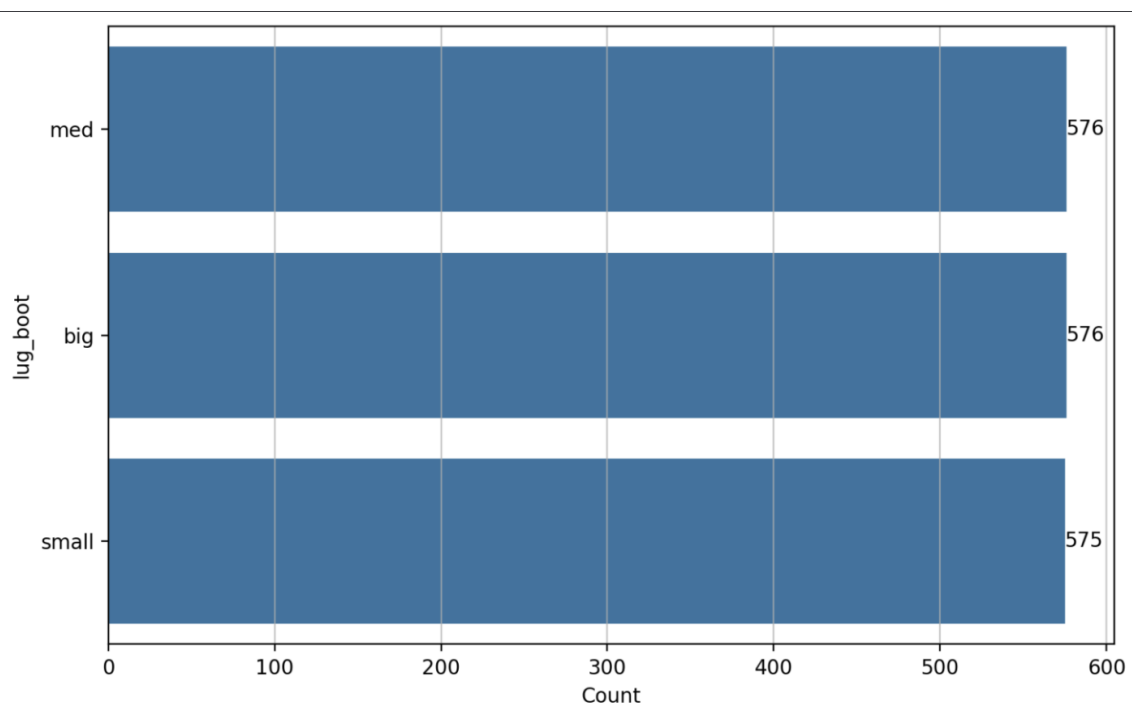


Figure 7 – Plotting Horizontal Bar Plots for 'lug_boot' Columns

The horizontal bar chart provides a visual summary of the distribution of the 'lug_boot' variable, which represents the size of the luggage boots in the dataset.

- Categories: The 'lug_boot' variable includes three categories "small", "med" (medium), and "big."
- Count: the "med" (medium) and "big" have an equal count of 576, the count of "small" (very high) is lower than the two categories above at 575. The x-axis displays the count, ranging from 0 to 600, while the y-axis lists the categories.

Analysis of 'safety' Variable

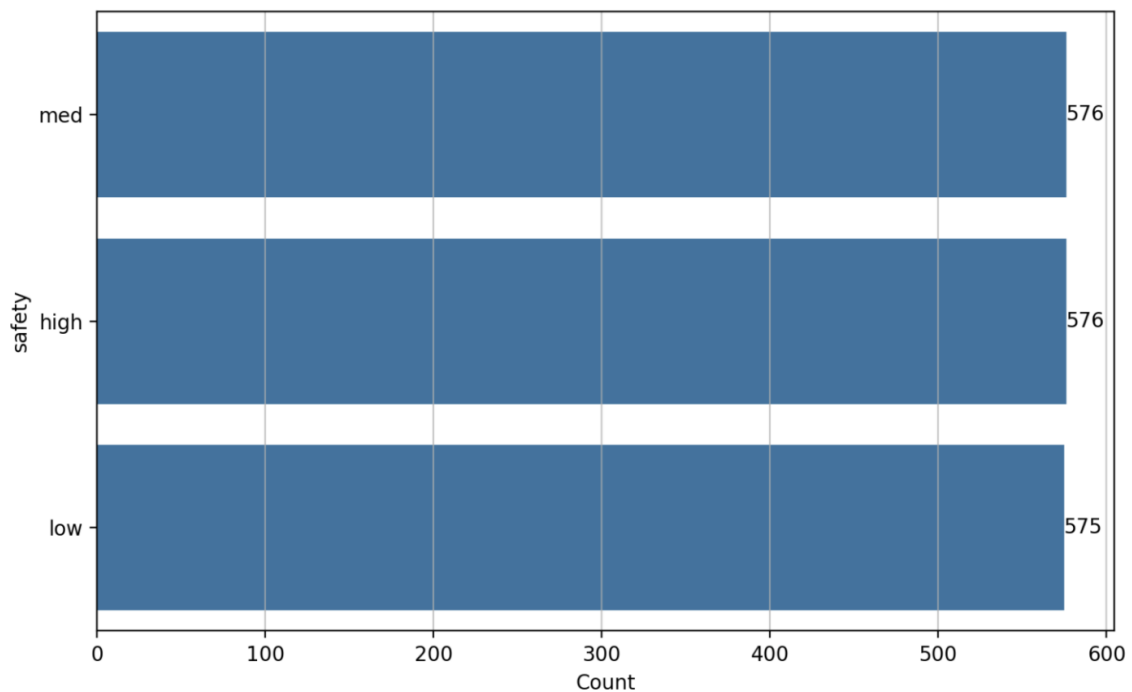


Figure 8 – Plotting Horizontal Bar Plots for 'safety' Columns

The horizontal bar chart provides a visual summary of the distribution of the 'safety' variable, which represents the estimated safety of the cars in the dataset.

- Categories: The 'safety' variable includes three categories "small", "med" (medium), and "high."
- Count: the "med" (medium) and "high" have an equal count of 576, the count of "low" (very high) is lower than the two categories above at 575. The x-axis displays the count, ranging from 0 to 600, while the y-axis lists the categories.

Analysis of 'Class' Variable

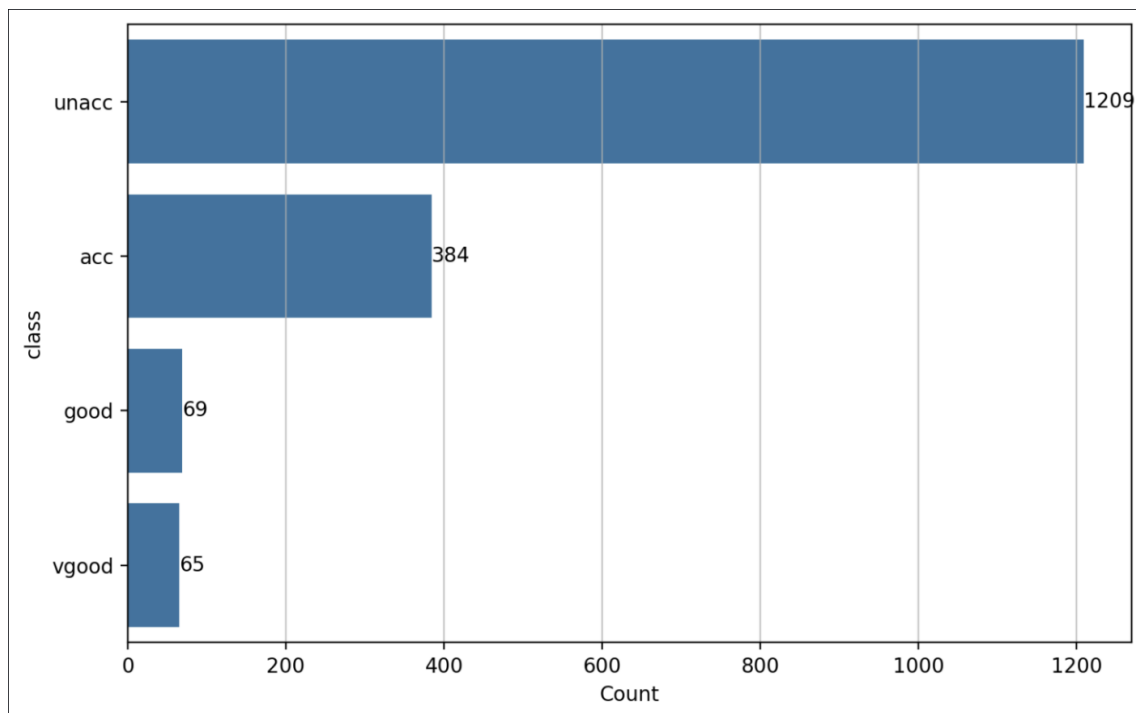


Figure 9 – Plotting Horizontal Bar Plots for 'class' Columns

The horizontal bar chart provides a clear visual representation of the distribution of the 'class' variable, which categorizes the evaluation level of the cars in the dataset:

- Categories: The 'maint' variable includes four categories "unacc" (unacceptable), "acc" (acceptable), "good", and "vgood" (very good).
- unacc: The "unacc" (unacceptable) category has the highest count, with 1210 instances. This dominance indicates that a significant majority of the cars in the dataset are evaluated as unacceptable.
- acc: The "acc" (acceptable) category comes next with 384 instances. This suggests a notable number of cars are deemed acceptable.
- good: The "good" category has 69 instances, indicating a smaller proportion of cars are evaluated positively.
- vgood: The "vgood" (very good) category has the lowest count, with 65 instances, signifying the rarest and most highly rated evaluations in the dataset.

Overall

- In general, almost all the categorical data in this dataset is evenly distributed. However, the 'class' data is unevenly distributed, with the largest number of 'unacc' ratings, this implies that many of the cars in the sample are viewed as unsatisfactory. In addition, no abnormalities were detected in the data.

Correlation matrix

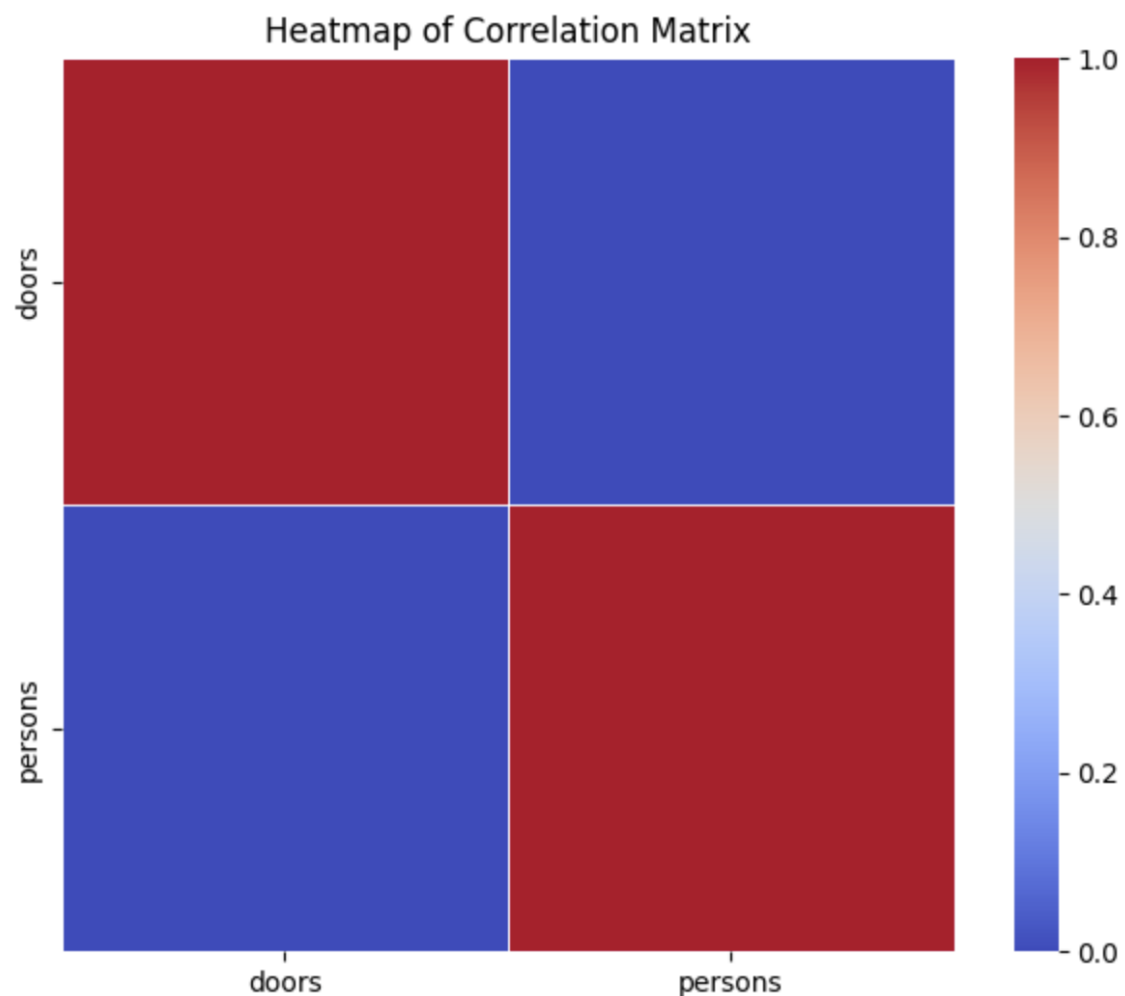


Figure 10 – Heatmap of Correlation Matrix

Purpose of the Heatmap:

- The heatmap visualizes the correlation matrix for five numeric columns: doors and person. The color scale represents the strength and direction of the correlation between each pair of variables, with red indicating strong positive correlations and blue indicating weaker correlations.

Analysis of Correlation Patterns:

Diagonal Elements:

- The diagonal elements (top-left and bottom-right) are red, indicating a perfect positive correlation (correlation coefficient of 1.0) between each variable and itself. This is expected as any variable is perfectly correlated with itself.

Off-Diagonal Elements:

- The off-diagonal elements (top-right and bottom-left) are blue, indicating a low or no correlation (correlation coefficient close to 0.0) between the variable's "doors" and "persons." This suggests that there is no linear relationship between the number of doors and the number of people.

Relevance:

- This heatmap is relevant for understanding the relationships between different variables in a dataset. In this case, it shows that the number of doors and the number of persons is not correlated.

Conclusion

- In conclusion, this dataset, derived from a straightforward hierarchical decision model, is ideal for experimenting with constructive induction and structure discovery techniques. It provides a comprehensive overview of car evaluations, featuring complete data, balanced categorical distributions, and clear numeric trends. These insights establish a strong foundation for further analysis, model building, and decision-making processes related to car evaluations.